Prism: Reducing Arithmetic for High-Recall Approximate Nearest Neighbor Search on Processing-in-Memory

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Vector Pruning
prunes distant vectors through dual-cluster affiliations

Reduced Computation
prioritizes high distance contribution vector segments with early termination

Host–PIM Pipeline
orchestrates host preparation, PIM execution, and global ranking

Background and Motivation

> **ANNS** fetches many high-dimensional vectors with random, fine-grained accesses and little temporal locality.

> **PIM** provides high internal bandwidth, but its on-die cores are compute-limited.

- **Arithmetic pressure:** exhaustive cluster scans can shift the bottleneck from bandwidth to PIM computation.
- **Recall preservation:** aggressive pruning or quantization risks losing high-recall results.
- **Coordination stalls:** host preparation, PIM execution, and global ranking must stay overlapped.

Contributions

- **Proximity-aware vector pruning** filters distant peripheral vectors before distance evaluation.
- **Sensitivity-ordered computation** evaluates high-impact dimensions first and safely terminates losers early.
- **Stall-free host–PIM pipeline** overlaps preparation, execution, and ranking across query batches.

System Overview

Three components: query profiler, distance calculation engine, and global ranker. Host-PIM pipeline coordinating both sides.

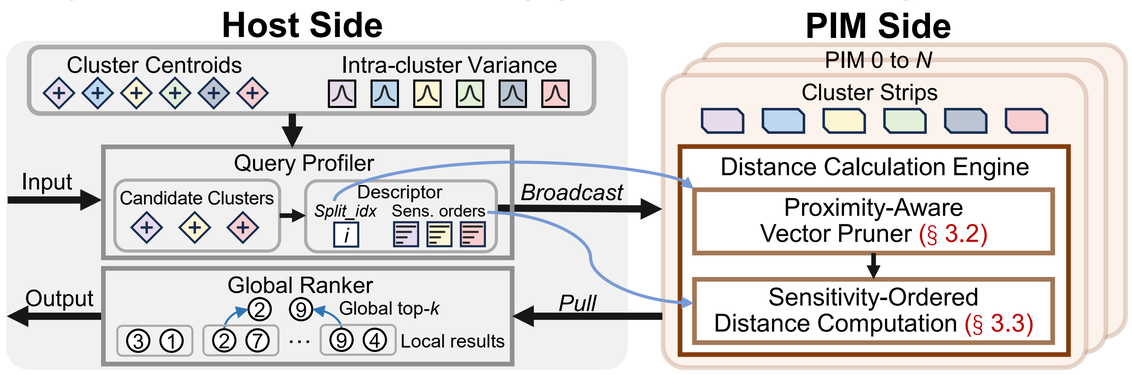


Fig. 1. Overview of Prism.

1. Proximity-Aware Vector Pruning

- **Offline:** k-means clustering + dual affiliation.
- **Online:** cluster partition and filtering.
- **Pruning:** exhaustively search central clusters and prune peripheral boundary vectors.

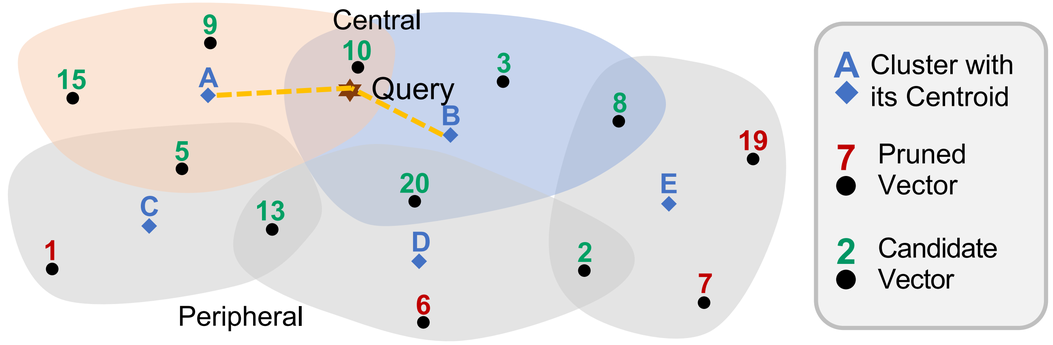


Fig. 2. Proximity-aware vector pruning.

2. Sensitivity-Ordered Computation

- **Host:** rank dimension segments by expected contribution.
- **PIM:** check each partial sum against the heap threshold.
- **Termination:** skips the remaining arithmetic once a vector cannot enter local results.

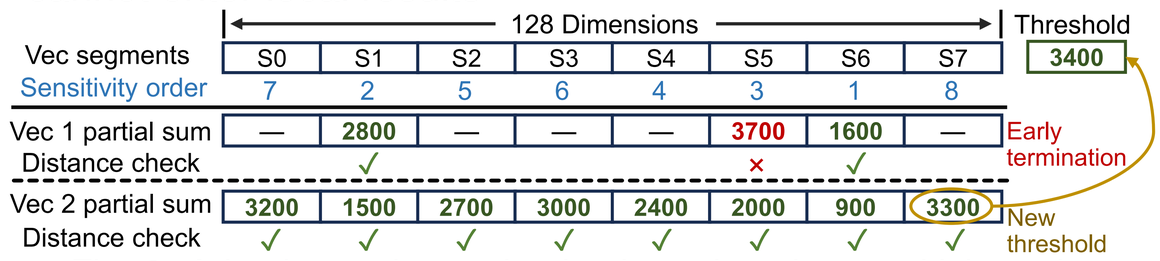


Fig. 3. Adaptive early termination based on the sensitivity order.

3. Stall-Free Host–PIM Pipeline

- **Prepare, execute, and rank** overlap host preparation, PIM execution, and global ranking.

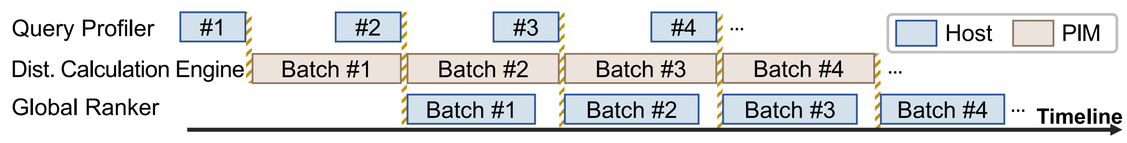


Fig. 4. Stall-free pipeline coordinating host and PIM.

Evaluation

Setup: 20 UPMEM DIMMs (2560 PIM modules); Sift100M (8 Segments) and SpaceV100M (10 Segments); batch size 128.

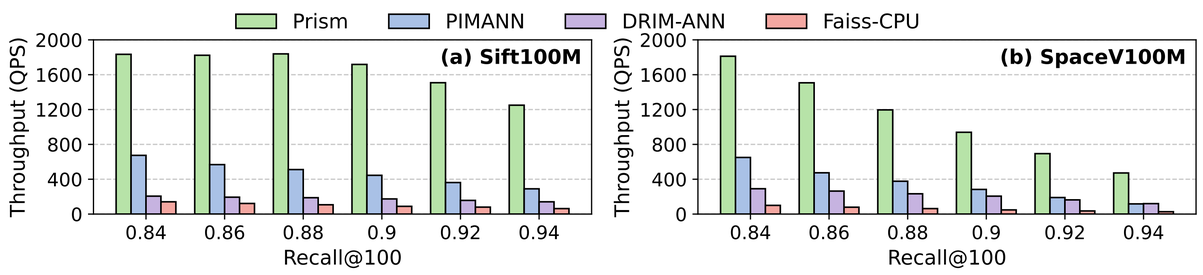


Fig. 5. Throughput under different recalls.

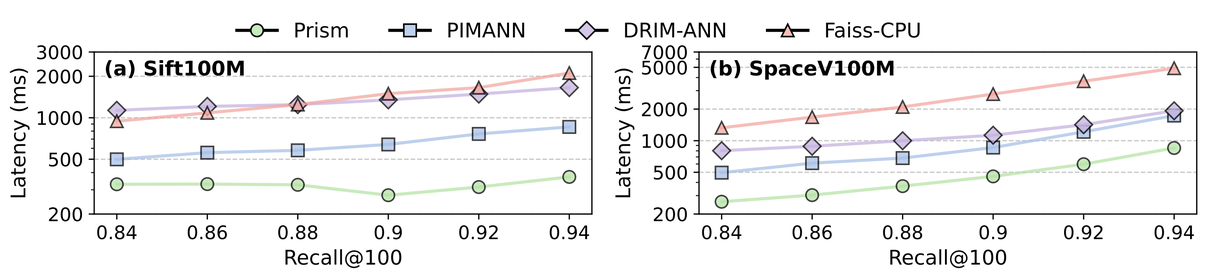


Fig. 6. Latency under different recalls (log scale).

13.0–19.8× over Faiss-CPU
3.9–9.9× over DRIM-ANN; 2.7–4.3× over PIMANN

65–84% lower than Faiss-CPU
56–80% lower than DRIM-ANN; 34–58% lower than PIMANN

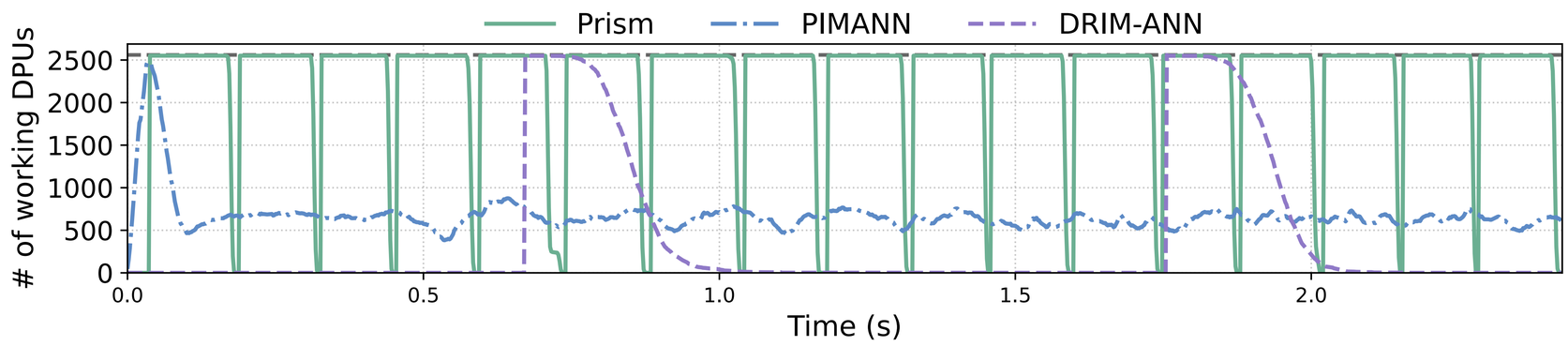


Fig. 7. The number of working DPUs of different systems.

Prism's working DPUs remain consistently high
eliminating intra-batch underutilization and inter-batch idleness

2.7–19.8×
higher throughput than state-of-the-art baselines

THE CHIPS TO SYSTEMS CONFERENCE
SHAPING THE NEXT GENERATION OF ELECTRONICS